TABLE I
COMPARISON WITH BASELINES (20 RUNS PER NON-DETERMINISTIC PLANNER)

	Cluttered environment								Powerlines environment							
			Best of the front								Best of the front					
Planner	SR (%)	Pl. Time (s)	Dur. (s)	Energy (kJ)	d_{min} (m)	d_{mean} (m)	HV	IGD$^+$	SR (%)	Pl. Time (s)	Dur. (s)	Energy (kJ)	d_{min} (m)	d_{mean} (m)	HV	IGD$^+$
ARENA (ours)	**100**	0.31 $\pm$.01	**254** $\pm$4	**802** $\pm$13	7.0 $\pm$.2	9.3 $\pm$.5	**0.93** $\pm$.02	**0.05** $\pm$.02	**100**	0.27 $\pm$.02	**151** $\pm$9	**495** $\pm$30	7.3 $\pm$.2	10.1 $\pm$1.6	**0.92** $\pm$.04	**0.05** $\pm$.03
MOAR-3D kino. sweep	**100**	15.4† (4.5)	338	1072	**7.1**	**11.3**	0.84	0.13	**100**	19.5† (17.0)	165	540	**7.5**	**10.6**	0.83	0.17
Fast-Planner [**?**]	**100**	**0.028** $\pm$.002	339	1075	1.3	6.0	0.17	0.59	**100**	**0.012**	170	558	1.8	6.5	0.18	0.83
Kino. RRT*	90	60.0	369 $\pm$8	1171 $\pm$26	1.6 $\pm$.5	6.2 $\pm$.6	0.25 $\pm$.10	0.51 $\pm$.09	**100**	60.0	206 $\pm$5	672 $\pm$15	2.7 $\pm$.9	6.5 $\pm$.5	0.28 $\pm$.16	0.69 $\pm$.21
A*	0	0.017 $\pm$.002	251	796	1.5	5.5	–	–	0	0.004 $\pm$.003	139	458	2.0	6.3	–	–
RRT*	0	0.91 $\pm$.01	270 $\pm$4	854 $\pm$14	1.6 $\pm$.4	5.6 $\pm$.5	–	–	0	0.75	186 $\pm$22	605 $\pm$69	2.3 $\pm$.9	6.4 $\pm$.8	–	–
MOAR-3D [**?**] sweep	0	0.36† (.60)	255	808	7.1	9.6	0.98	0.01	0	0.22† (.25)	147	484	7.5	9.8	0.95	0.03
Ahmed and Deb [**?**]	0	1.51 $\pm$.08	248 $\pm$6	783 $\pm$19	4.3 $\pm$1.1	8.0 $\pm$.8	0.79 $\pm$.13	0.10 $\pm$.09	0	0.94 $\pm$.04	152 $\pm$11	496 $\pm$34	6.5 $\pm$1.2	10.4 $\pm$1.3	0.85 $\pm$.07	0.14 $\pm$.07
A*pex [**?**]	0	23.9 $\pm$.5	255	808	7.1	11.9	0.98	0.02	0	103 $\pm$2	147	484	7.5	12.4	0.94	0.03
PO-RRT* [**?**]	0	5.01	311 $\pm$13	986 $\pm$41	1.9 $\pm$.4	6.3 $\pm$.8	0.35 $\pm$.11	0.41 $\pm$.10	0	5.01	218 $\pm$46	707 $\pm$143	4.2 $\pm$1.2	8.7 $\pm$.9	0.48 $\pm$.17	0.47 $\pm$.23

SR: success rate. Pl. Time: planning time. Dur.: duration. d_{min}, d_{mean}: minimum and mean clearance. HV: hypervolume. IGD$^+$: inverted generational distance plus. Gray: trajectories violate constraints. † median (IQR) shown instead of mean $\pm$ std because of skewed distribution.